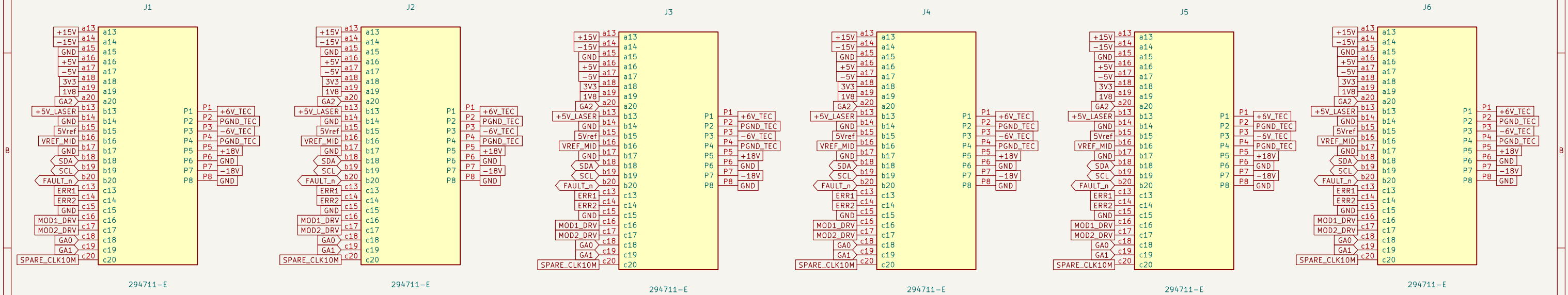


UNIVERSAL SLOT – every backplane position wired identically (DIN 41612 Type M 24+8)

any card in any slot; slot identity via GA2..GA0 straps on the backplane



MECHANICAL – SLOT PITCH (2026–07–15):

Connector pitch = 10 HP = 50.8 mm centre-to-centre, UNIFORM across all 8 slots
(any-card-any-slot requires one pitch). 8 x 10HP = 80HP -> standard 84HP 3U subrack, 4HP spare.
Tallest components above PCB (measured from footprints/descriptions):
39.1 mm GBPC bridge (7.3) + Boyd 6223BG sink (31.75) – pm22V/pm6V <- sets the pitch
38.1 mm AAVID 531102B extruded (pm6V/pm22V/pm1p8V): ~35 mm D18 radial caps (typ, VERIFY)
25.4 mm Wakefield 657–10ABPE (pm6V; –10 variant chosen to fit eurocard)
Budget at 10HP: 50.8 – 1.6 PCB – 2.5 neighbour solder-side = 46.7 mm envelope
-> 7.6 mm air over tallest sink (natural-convection rated; fins vertical = chimney per bay).
9HP (45.72) is the bare minimum but chokes convection (~2.5 mm gap) – only with a fan tray.
8HP (40.64) DOES NOT FIT. Caliper-check 6223BG stack + D18 cap heights on fabbed boards.

GA STRAPPING (per backplane slot; 0 = GND, 1 = 3V3):

slot 1: GA2=0 GA1=0 GA0=0 (e.g. pm22V – ignores GA)
slot 2: GA2=0 GA1=0 GA0=1 (e.g. pm6V – ignores GA)
slot 3: GA2=0 GA1=1 GA0=0 (e.g. pm1p8V – ignores GA)
slot 4: GA2=0 GA1=1 GA0=1 (e.g. DIG – I2C master)
slot 5: GA2=1 GA1=0 GA0=0 (e.g. AFE)
slot 6: GA2=1 GA1=0 GA0=1 (e.g. CH: GA0=1 -> laser 1 nets)
slot 7: GA2=1 GA1=1 GA0=0 (e.g. CH: GA0=0 -> laser 2 nets)
slot 8: GA2=1 GA1=1 GA0=1 (spare)

PER-CARD PIN USAGE (all cards carry the same connector):

pm22V (FABBED): P5–P8 only. pm6V (FABBED): P1–P4 only. Both slot-agnostic as built.
pm1p8V: drives rails a13–b17; draws +/-18V from P5/P7.
DIG: I2C master (SDA/SCL); drives MOD1/2 from DDS; ADC-listens ERR1/2;
monitors all rails incl. blades; FAULT_n pull-up + read; front: USB, TFT, 10MHz SMA, LO SMAs.
CH x2 (identical): GA0 selects ERR1/MOD1 vs ERR2/MOD2 via on-card ADG419;
local I2C DAC (setpoints) + I2C ADC (IMON/NTC/MPD) at GA-derived address;
TEC power from P1–P4 (returns ONLY on PGND_TEC); laser head conn on front panel.
AFE: drives ERR1 + ERR2; local I2C ADC for PD monitors; LO1/LO2 in via front SMA;
fiber FC receptacles on front panel.

DESIGN RULES / NOTES (draft 2026–07–15):

- Rail bus a13–b17 + blades P1–P8 match pm1p8V/pm6V/pm22V as built – fabbed boards unchanged.
- TEC current returns ONLY on P2/P4 (PGND_TEC); joined to GND at single star point at pm6V output.
- I2C: 400 kHz max, pull-ups on DIG card only; 100R series in SDA/SCL at every card entry;
bus is STATIC during lock (quiet mode) – poll telemetry <= 10 Hz, never during acquisition.
- Keep dither/mod frequency away from N x 100/400 kHz (I2C harmonics fold through PDH demod).
- ERR1/ERR2: single driver (AFE), listeners = CH + DIG. MOD1/2: single driver (DIG).
- GND at c15 guards ERR/MOD group; GA straps are DC – safe next to analog.
- FAULT_n: open-drain wired-OR, any card may assert; DIG queries source over I2C.
- LO and 10 MHz distribution via front-panel coax; c20 SPARE_CLK10M is fallback only.
- On CH/AFE cards: RC-filter DAC outputs (<=1 kHz), 10R+10uF on I2C–chip 3V3,
digital cluster in the connector corner, away from summing nodes / sense amps.

Setpoints/monitors digitized per-card over shared I2C; ERR/MOD analog bused
One Type M 24+8 pinout for every slot; GA straps give slot identity

Sheet: /
File: Backplane_Slots.kicad_sch
Title: v5 backplane UNIVERSAL slot pinout (DRAFT rev B)
Size: A3 Date: 2026–07–15 Rev:
KiCad E.D.A. 10.0.3 Id: 1/1